rSoccer Intro

1 Introduction

rSoccer is an open-source simulator for the IEEE Very Small Size Soccer and the Small Size League optimized for reinforcement learning (RL). It provides task environments with a field, robots and a ball, and uses rSim to simulate the physics. In this lab course, we provide you with an API for rSoccer that lets you design, simulate, and test autonomous robot soccer strategies in a simulated environment.

For more information about the simulator, read the paper by Martins et al. (2022) [1].

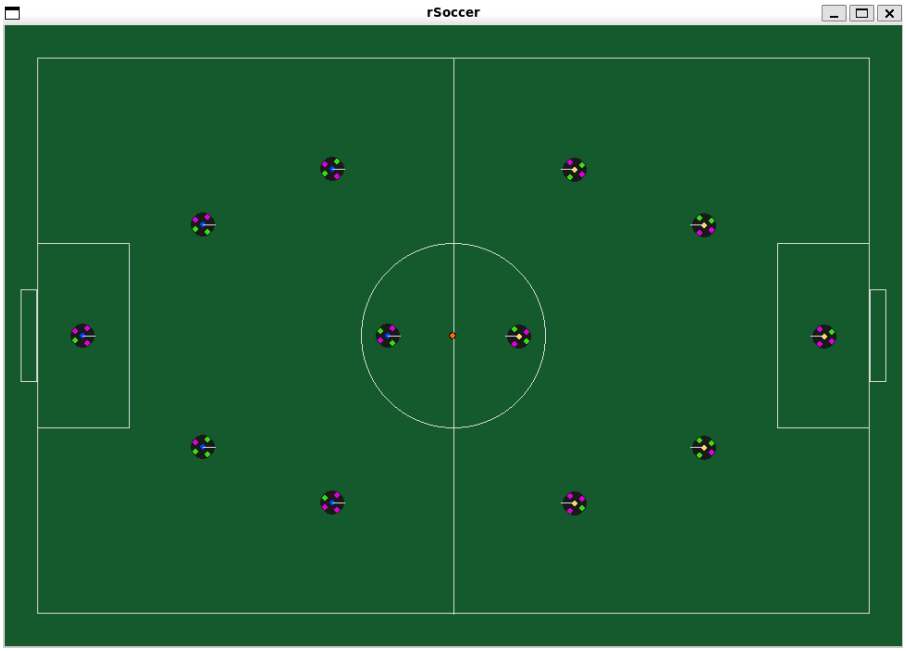


Figure 1: Screenshot of the rSoccer simulator.

2 Installing rSoccer

The official installation instructions are available at rSoccer repository. Do not install through the Python Package Index (PyPI) as you might face dependency issues.

Though, We **strongly recommend** installing rSoccer alongside the API by following the instructions in the API repository's README. This approach simplifies the installation process and sets up a virtual environment to keep project dependencies isolated, preventing conflicts with other Python packages and ensuring everything runs smoothly.

3 Running rSoccer

First activate the virtual environment you used to install rSoccer. Then run the following code as an example:

```
import gymnasium as gym
import numpy as np
import pygame


env = gym.make("SSLStaticDefenders-v0", render_mode="human")
observation, _ = env.reset()

while not pygame.event.get([pygame.QUIT, pygame.KEYDOWN]):
    ball_x, ball_y = observation[0], observation[1]
    robot_x, robot_y = observation[4], observation[5]
    robot_angle = np.arctan2(observation[6], observation[7])

    to_ball_x, to_ball_y = ball_x - robot_x, ball_y - robot_y
    turn_angle = np.arctan2(to_ball_y, to_ball_x) - robot_angle

    action = [to_ball_x, to_ball_y, turn_angle, 1, 0]
    observation, _, terminated, truncated, _ = env.step(action)
    if terminated or truncated:
        for _ in range(20):
            env.step([0, 0, 0, 0, 0])
        observation, _ = env.reset()

env.close()
```

The robot moves towards the ball and kicks it. The scene then restarts. To stop, close the window or press any keys on your keyboard.

4 rc-robosim Error on Windows

If you face an rc-robosim error during installation, the reason is rc-robosim has to be compiled on Windows and cannot find the ODE physics library. Make sure Visual Studio Build Tools is installed, then run the following commands in PowerShell (not Command Prompt) inside your activated environment:

```
conda install -c conda-forge libode
$p = $env:CONDA_PREFIX -replace '\\','/'
$env:CMAKE_ARGS = "-DODE_INCLUDE_DIRS=$p/Library/include"
$env:CMAKE_ARGS += " -DODE_LIBRARY_RELEASE=$p/Library/lib/ode_double.lib"
$env:CMAKE_ARGS += " -DCMAKE_POLICY_VERSION_MINIMUM=3.5"
pip install .
```

5 Setting Up WSL

If you still encounter issues installing rc-robosim (rSoccer) directly on Windows, you can use Windows Subsystem for Linux (WSL2) as an alternative. WSL2 allows you to run a Linux

environment on Windows without setting up a separate virtual machine. Use the following links for installation guide:

WSL2 Installation Guide
VS Code with WSL2
PyCharm with WSL2

6 Setting Up the Virtual Machine and Ubuntu

Alternatively, you can install rSoccer in a virtual machine (VM) running Ubuntu 24.04. We use Oracle VirtualBox to create and run the VM.

- From the download page, under **VirtualBox Platform Packages**, choose and download the installer matching your operating system.
- Install Oracle VirtualBox using the downloaded package. Full details at User Guide.
- Now download Ubuntu 24.04 Desktop image (for Apple Silicon Macs).
- After the download is complete, open Oracle Virtual box but before creating a new VM it is better to have a look at this web page.
- After opening the Oracle Virtual box click on the **New** icon. Choose a name for your virtual machine, and in **ISO Image** select the downloaded Ubuntu `.iso` file and Install Ubuntu in your virtual machine. Full details at User Guide

7 rSoccer ball rendering

rSoccer renders the environment so that you are able to observe your strategies while the simulation is running. However, the ball is really small and it can be difficult to be spotted. For this reason we provide you with a file that will render a white cross on top of the ball so that it is easier to be spotted. The new file that you can find in this folder is called *ball.py* and you should use it to replace the file inside the following rsoccer folder *rsoccer_gym/Render/ball.py*.

References

[1] F. B. Martins, M. G. Machado, H. F. Bassani, P. H. M. Braga, and E. S. Barros, "rSoccer: A Framework for Studying Reinforcement Learning in Small and Very Small Size Robot Soccer," in *RoboCup 2021: Robot World Cup XXIV*, pp. 165–176, Springer, Cham, 2022.